EXPERIMENT 16

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

Exploratory Data Analysis – Quantitative and Qualitative

Aim: Perform numerical and categorical analysis and visualize relationships.

Python Program

```
import seaborn as sns
import matplotlib.pyplot as plt
df=sns.load_dataset('tips')
print(df.describe())
print(df.describe(include='all'))
sns.histplot(df['total_bill'],kde=True); plt.show()
sns.boxplot(x='day',y='total_bill',data=df); plt.show()
sns.countplot(x='sex',data=df); plt.show()
sns.heatmap(df.corr(numeric_only=True),annot=True); plt.show()
```

Result

Statistics and plots summarize numerical variables, categories and correlations.

Note: For CSV programs, keep the named CSV file in the same folder and check that column names match.